YIXUAN "CONNIE" ZHU

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EDUCATION

University of Southern California — Los Angeles, CA
M.S. in Computer Science — Jan 2025 – Dec 2026

Rose-Hulman Institute of Technology — Terre Haute, IN
B.S. in Computer Science — Sep 2019 – Nov 2023

WORK EXPERIENCE

Ingram Micro — Jun 2026 – Present
AI & LLM Engineering Intern

- Developed an AI agent that conducts agent-to-agent (A2A) attacks, independently probing other AI agents for weaknesses that traditional testing often misses
- Pioneered chain-of-thought attack techniques that target a model's step-by-step reasoning process, uncovering vulnerabilities invisible to standard prompt-based testing
- Identified 12 vulnerabilities, including 3 rated high-severity, enabling the team to resolve high-risk issues prior to production release
- Reduced manual red-teaming effort by 86% and delivered prioritized risk reports that translated technical findings into actionable recommendations

LLM & GENERATIVE AI PROJECTS

Instruction-Guided Sparse Code Editing with Diffusion Language Models — Sep 2026 - Present

- Researching non-autoregressive code generation with Diffusion Language Models to perform instruction-guided bug fixing without explicit edit masks or predefined modification boundaries.
- Exploring instruction-conditioned editing localization across source-code lines and insertion gaps to support sparse code modifications through iterative masked-token denoising.
- Benchmarked diffusion-based code editing against full-file regeneration and oracle-localized infilling to investigate localization bottlenecks, generation capacity, functional correctness, and unintended code changes.

Knowledge Compression with User Feedback for RAG — Sep 2025 – Mar 2026

- Architected an end-to-end RAG pipeline using GPT models to auto-generate 62K high-quality QA pairs for retrieval-system evaluation
- Developed clustering-based knowledge-compression algorithms that reduced knowledge-base size by 80% while improving weighted recall by 6.8% and accuracy by 10.9%
- Trained custom Word2Vec embeddings on agent-generated QA sequences and implemented vector indexing for scalable deployment

LLM Fine-Tuning Pipeline — Mar 2025

- Fine-tuned LLaMA 3 (8B) with Llama Factory (SFT) on 50K domain-specific samples, applying LoRA adapters and mixed-precision training to cut GPU training time by 40%
- Built a scalable preprocessing pipeline (JSONL conversion, augmentation, batching, tokenization) and containerized inference with Docker/FastAPI, reducing average latency by 30%
- Developed an evaluation framework (BLEU/ROUGE, perplexity, human-in-the-loop) that improved response relevance and coherence by 15%

Multi-Agent LLM System — Dec 2025

- Built a local search agent with LangChain and custom tool integrations for real-time information retrieval
- Orchestrated specialized CrewAI agents to automate marketing-strategy generation and complex business workflows

TECHNICAL SKILLS

LLM & GenAI: LLaMA, GPT, Transformer architectures, LoRA/QLoRA fine-tuning, prompt engineering, RAG pipelines, vector databases (FAISS, Pinecone), embeddings (Word2Vec, Sentence Transformers)
Frameworks & Tools: LangChain, CrewAI, Llama Factory, Hugging Face Transformers, PyTorch, TensorFlow, FastAPI, Docker
NLP & ML: Text classification, NER, sentiment analysis, speech processing, TF-IDF, NLTK, scikit-learn
Languages: Python, SQL, Java, C/C++, JavaScript, R, MATLAB
Infrastructure: Linux/Ubuntu, Git, Google Colab, Jupyter, mixed-precision training, model quantization

RESEARCH EXPERIENCE

Shenzhen Research Institute of Big Data — Shenzhen, China
Research Assistant Intern — Jul 2023 – Aug 2023

- Built a target-speaker-extraction framework on the open-source WeSpeaker project, integrating voice activity detection (VAD) for data shuffling and segmentation
- Tuned Conv-TasNet and BSRNN models to improve extraction accuracy and computational efficiency

Rose-Hulman Institute of Technology — Terre Haute, IN
Undergraduate Researcher — Sep 2022 – Nov 2023

- Augmented speech datasets with Librosa noise profiles and evaluated their impact on Conv-TasNet and DNN-based speech-enhancement models

PUBLICATIONS

Zhu, Y., Ratnakar, S., Cheng, C., & Vijayakumar, C. "One Retrieval to Cover Them All: Co-occurrence-Aware Knowledge Base Reorganization for Session-Level RAG," KnowFM @ ACL, Apr 2026.

Zhu, Y., & Olga, S. "Speech Enhancement: Data Manipulation Techniques for Augmenting Existing Datasets," Springer LNNS, Nov 2024.